

Fake Job Posting Detection — Model Performance Report

Seven experiments using the EMSCAD dataset (fake_job_postings.csv, 17,880 postings, 4.8% fraudulent) after the shared Data Cleaning and Feature Engineering steps. All experiments use an 80/20 stratified train/test split with random_state = 42.

Accuracy note: Accuracy for EX1-EX2 is approximate, derived from the reported Precision/Recall and the test-set class distribution. Accuracy for the remaining experiments is preserved from the supplied report.

EX1 — One-Hot Encoding + Numerical Features

Setup: Only categorical and numerical features were used, without text features. Categorical features used One-Hot Encoding, while numerical features used StandardScaler. Six models were evaluated without hyperparameter tuning.

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
Random Forest	0.964	0.618	0.754	0.956	≈ 0.981
Decision Tree	0.708	0.699	0.703	0.842	≈ 0.981
XGBoost	0.912	0.538	0.676	0.955	≈ 0.975
Logistic Regression	0.225	0.809	0.352	0.912	≈ 0.856
Linear SVC	0.219	0.792	0.343	0.902	≈ 0.853
Complement NB	0.144	0.763	0.242	0.856	≈ 0.769

Best Models: Best F1 — Random Forest (0.754). Best Recall — Logistic Regression (0.809). Best Precision — Random Forest (0.964). Best ROC-AUC — Random Forest (0.956).

Conclusion: Random Forest gave the strongest overall performance, with F1 = 0.754 and Precision = 0.964. Logistic Regression achieved the highest Recall (0.809), but its low Precision reduced its F1-score.

EX2 — Ordinal Encoding + Numerical Features

Setup: The same categorical and numerical features were used, but categorical features were encoded with Ordinal Encoding instead of One-Hot Encoding. The models were evaluated without hyperparameter tuning.

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
Random Forest	0.936	0.595	0.728	0.950	≈ 0.977
XGBoost	0.889	0.555	0.683	0.963	≈ 0.973
Decision Tree	0.636	0.647	0.642	0.814	≈ 0.964
Linear SVC	0.160	0.613	0.253	0.797	≈ 0.940
Logistic Regression	0.154	0.624	0.247	0.797	≈ 0.938
Complement NB	0.080	0.584	0.141	0.673	≈ 0.923

Best Models: Best F1 — Random Forest (0.728). Best Recall — Decision Tree (0.647). Best Precision — Random Forest (0.936). Best ROC-AUC — XGBoost (0.963).

Conclusion: Random Forest remained best, but F1 dropped from 0.754 (One-Hot) to 0.728 (Ordinal); linear models degraded considerably. One-Hot Encoding is preferred; Ordinal is kept only as a comparison.

EX3 — Classical Machine Learning + TF-IDF

Setup: Six classical models combined TF-IDF features from job_context, One-Hot Encoded categorical features, and numerical / information-availability features.

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
Logistic Regression	0.706	0.734	0.720	0.946	≈ 0.972
Decision Tree	0.745	0.694	0.719	0.841	≈ 0.974
Linear SVC	0.662	0.769	0.711	0.958	≈ 0.970
Random Forest	0.990	0.549	0.706	0.975	≈ 0.978
XGBoost	0.941	0.549	0.693	0.966	≈ 0.977
Complement Naive Bayes	0.445	0.746	0.557	0.923	≈ 0.943

Conclusion: Logistic Regression had the highest F1 (0.720); Linear SVC had the highest Recall (0.769). F1 is lower than EX1, so job_context alone was not enough to improve results.

EX4 — Text Context + Numerical Features

Setup: title, description, requirements, and benefits were combined into one text_combined feature, transformed with TF-IDF, and paired with numerical / information-availability features.

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
Logistic Regression	0.842	0.803	0.822	0.983	≈ 0.983
Linear SVC	0.879	0.757	0.814	0.979	≈ 0.983
XGBoost	0.981	0.607	0.750	0.978	≈ 0.980
Random Forest	0.812	0.624	0.706	0.976	≈ 0.975
Decision Tree	0.679	0.723	0.700	0.862	≈ 0.970
Complement Naive Bayes	0.458	0.884	0.604	0.973	≈ 0.944

Conclusion: Clear improvement over EX3 (F1: 0.720 → 0.822). Logistic Regression had the best F1 (0.822) and ROC-AUC (0.983) — a richer text representation was clearly beneficial.

EX5 — Imbalance Handling

Setup: The dataset is highly imbalanced (4.8% fraud). Random Over-Sampling (ROS) and SMOTE were applied to the training data only, on top of the EX4 text + numerical feature setup.

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
Linear SVC + SMOTE	0.893	0.769	0.826	0.980	≈ 0.984
Logistic Regression + SMOTE	0.866	0.786	0.824	0.984	≈ 0.984
Logistic Regression + ROS	0.866	0.786	0.824	0.985	≈ 0.984
Linear SVC + ROS	0.892	0.763	0.822	0.980	≈ 0.984
XGBoost + SMOTE	0.876	0.694	0.774	0.962	≈ 0.980
Random Forest + SMOTE	1.000	0.601	0.751	0.974	≈ 0.981
Random Forest + ROS	0.707	0.751	0.728	0.980	≈ 0.973
XGBoost + ROS	0.580	0.821	0.679	0.974	≈ 0.963
Decision Tree + ROS	0.636	0.688	0.661	0.834	≈ 0.966

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
Decision Tree + SMOTE	0.594	0.728	0.655	0.852	≈ 0.963

Conclusion: Linear SVC + SMOTE had the highest F1 (0.826). Random Forest + SMOTE reached perfect Precision (1.000) but Recall was only 0.601, so it is not the best overall choice.

EX6 — Ensemble Learning

Setup: Hard Voting and Stacking were evaluated using the strongest models from earlier experiments.

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
Stacking	0.950	0.769	0.850	0.983	0.987
Logistic Regression + ROS	0.866	0.786	0.824	0.985	0.984
Hard Voting	0.992	0.676	0.804	—	0.984

Conclusion: Stacking is the best model in the project so far — F1 = 0.850, Precision = 0.950, Recall = 0.769, ROC-AUC = 0.983, Accuracy = 0.987. Hard Voting had the highest Precision (0.992) but lower Recall (0.676), so Stacking offers the best overall balance.

EX7 — DistilBERT Feature Extraction

Setup: A pre-trained DistilBERT model was used strictly as a frozen feature extractor (not fine-tuned). Job text → tokenizer → frozen DistilBERT → mean-pooled embeddings → combined with numerical features → XGBoost. Best configuration used a tuned decision threshold of 0.22.

Model	Precision	Recall	F1-Score	ROC-AUC	Accuracy
DistilBERT + XGBoost	0.881	0.688	0.773	0.982	0.980

Conclusion: DistilBERT gave strong results (F1 = 0.773) but did not outperform the Stacking ensemble (F1 = 0.850). The ensemble approach was better on this dataset.

Overall Comparison — All 7 Experiments

Exp	Best Model	F1	Recall	Precision	ROC-AUC	Accuracy
EX1	Random Forest	0.754	0.618	0.964	0.956	≈ 0.981
EX2	Random Forest	0.728	0.595	0.936	0.950	≈ 0.977
EX3	Logistic Regression	0.720	0.734	0.706	0.946	≈ 0.972
EX4	Logistic Regression	0.822	0.803	0.842	0.983	≈ 0.983
EX5	Linear SVC + SMOTE	0.826	0.769	0.893	0.980	≈ 0.984
EX6	Stacking	0.850	0.769	0.950	0.983	0.987
EX7	DistilBERT + XGBoost	0.773	0.688	0.881	0.982	0.980

Final Conclusion: Seven experiments compared structured-feature encoding, text-based classical ML, imbalance handling, ensemble learning, and transformer-based feature extraction. The best overall result was achieved by Stacking in EX6 (F1 = 0.850, Accuracy = 0.987). One-Hot Encoding was preferable to Ordinal Encoding. DistilBERT was used only as a frozen feature extractor and did not outperform the ensemble approach on this dataset.